Exercise 9.6

1. (i) $\frac{63\pi}{512}$ (ii) $\frac{16}{35}$ (iii) $\frac{5\pi}{64}$ (iv) $\frac{8}{45}$ (v) $\frac{\pi}{32}$ (vi) $\frac{64}{35}$ (vii) $\frac{1}{24}$ (viii) $\frac{1}{60}$

Exercise 9.7

1. (i) $\frac{5!}{3^6}$ (ii) 29 (2) $\frac{1}{8}$

Exercise 9.8

1. 7.5 2. 2 3. 15 4. 36 5. $2\sqrt{2}$ 6. $\log 2$ 7. $\frac{9}{2}$ 8. yes, $\frac{16}{3}$ 9. $\frac{4}{3}$ 10. $\frac{4}{3}\left(4\pi+\sqrt{3}\right)$

Exercise 9.9

1. $\frac{4\pi}{5}$ 2. $\frac{\pi}{4}\left[1-e^{-4}\right]$ 3. 8π 4. $\frac{2\pi}{15}$ 5. $\frac{14}{3}\pi m^3$ 6. $\frac{1000}{3}\pi cm^3$

Exercise 9.10

1	2	3	4	5	6	7	8	9	10
(1)	(3)	(3)	(4)	(4)	(3)	(3)	(3)	(2)	(1)
11	12	13	14	15	16	17	18	19	20
(4)	(2)	(2)	(4)	(4)	(4)	(3)	(4)	(2)	(1)

Chapter 10

Exercise 10.1

1. (i) 1,1 (ii) 3,2 (iii) 2, does not exist (iv) 1, 2 (v) 1,4

(vi) 2,2 (vii) 2,6 (viii) 2, does not exist (ix) 3,1 (x) 1, 1

Exercise 10.2

1. (i) $\frac{dQ}{dt}=kQ$ (ii) $\frac{dP}{dt}=kP(500000-P)$ (iii) $\frac{dP}{dT}=\frac{kP}{T^2}$ (iv) $\frac{dx}{dt}=\frac{2x}{25}+400$ 2. $\frac{dr}{dt}=-k$

Exercise 10.3

1. (i) $\frac{d^2y}{dx^2}=0$ (ii) $\frac{d^2x}{dy^2}=0$ 2. $r^2\left[1+\left(\frac{dy}{dx}\right)^2\right]=\left(x\frac{dy}{dx}-y\right)^2$

3. $x^2+2xy\frac{dy}{dx}-y^2=0$ 4. $2ay''+y'^3=0$ 5. $xy'-2y-2=0$ 6. $xy'^2+xyy''-yy'=0$

7. $\frac{d^2y}{dx^2}=64y$ 8. $xy''+2y'+x^2-xy-2=0$

Exercise 10.4

2. (i) $m=-2$ (ii) $m=2,3$ 3. $2y^2=x+48$

Exercise 10.5

1. $F=(F-kV)e^{\frac{kt}{M}}$ 2. $k^2\left(1-e^{-\frac{2gx}{k^2}}\right)=v^2$ 3. $y=\frac{1-x}{1+x}$

4. (i) $\sin^{-1}y=\sin^{-1}x+C$ (ii) $y\tan^{-1}x=C$ (iii) $\sin\left(\frac{y-1}{x}\right)=a$ (iv) $e^x+e^{-y}+\frac{x^4}{4}=C$

(v) $\left(e^y+1\right)\sin x=C$ (vi) $\sin\left(\frac{x}{y}\right)=e^{nx+c}$ (vii) $3y=-\left(25-x^2\right)^{\frac{3}{2}}+3C$ (viii) $\sin y=e^x\log x+C$

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